15th IEEE Workshop on Hyperspectral Imaging and Signal Processing: Evolution in Remote Sensing (WHISPERS)
Barcelona, Spain, Nov 11 – Nov 14, 2025

Hyperspectral Image Classification using Spectral-Spatial Mixer Network

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Paper ID: 12

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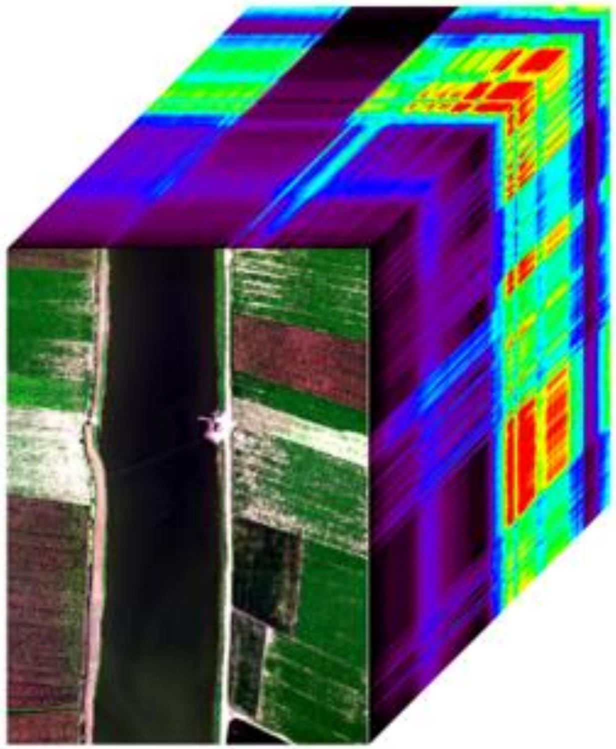


Motivation

- Existing CNN and Transformer models struggle to efficiently capture both local spectral–spatial patterns and global dependencies in hyperspectral images.

- Transformer-based methods achieve high accuracy but depend on large labeled datasets and high computational resources, limiting their practicality in remote sensing.

- A compact and effective model is needed to deliver robust classification performance with reduced computational cost.

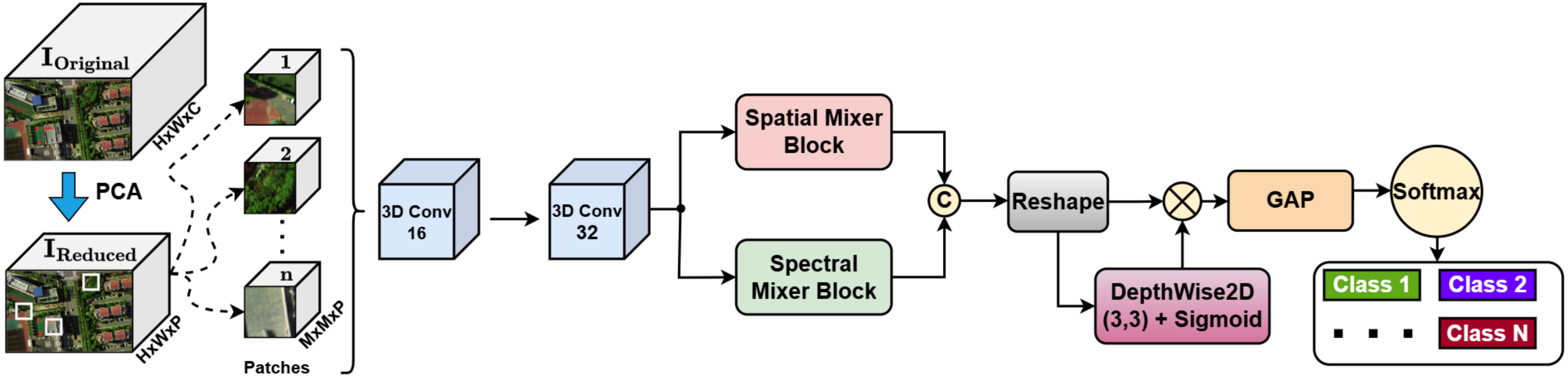
Contributions

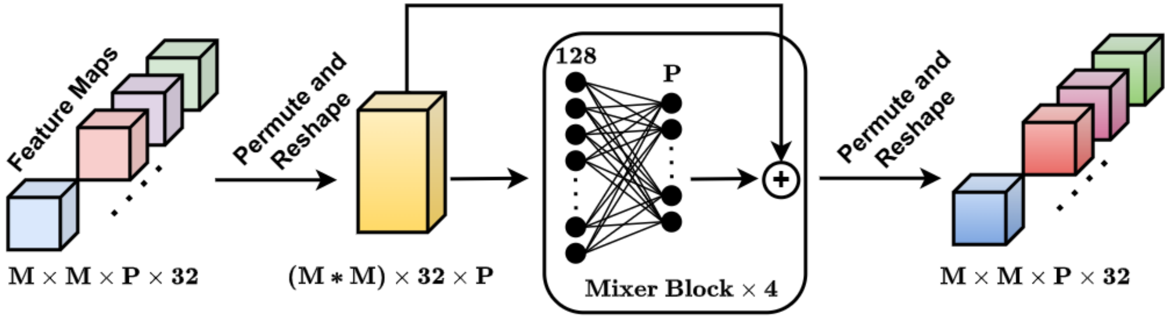
- A lightweight architecture (SS-MixNet) is introduced, combining 3D convolutional layers with spectral and spatial MLP mixer blocks for joint spectral–spatial feature learning.

- A depthwise convolution-based attention mechanism is integrated to enhance feature discrimination with minimal computational overhead.

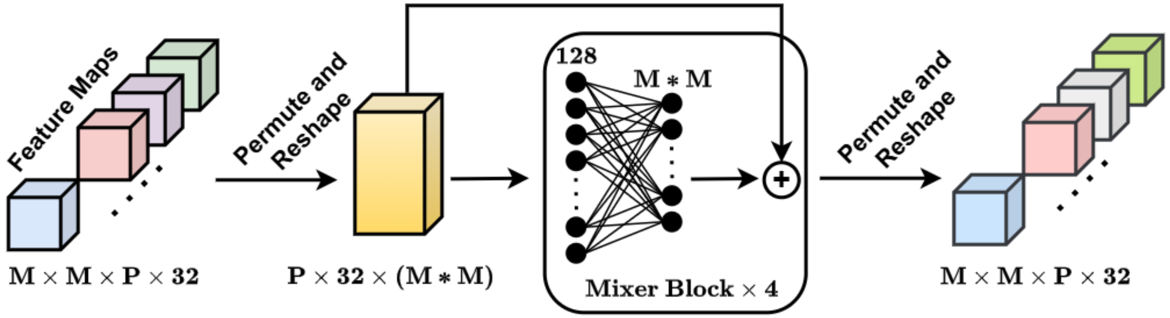
- The model achieves state-of-the-art accuracy on QUH-Tangdaowan and QUH-Qingyun datasets using only 1% labeled data, demonstrating strong efficiency and robustness.

Methodology – Proposed Model



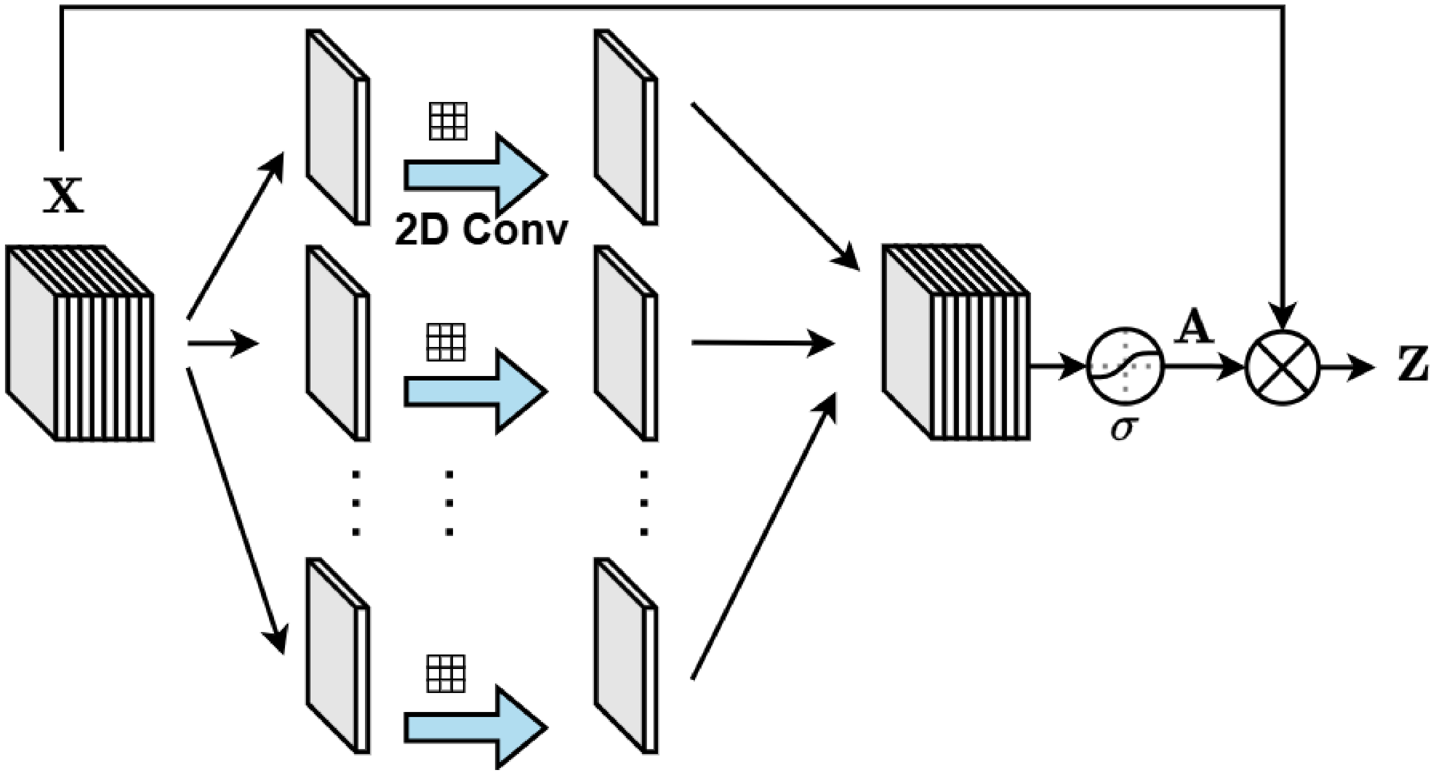


Spectral Mixer Block



Spatial Mixer Block

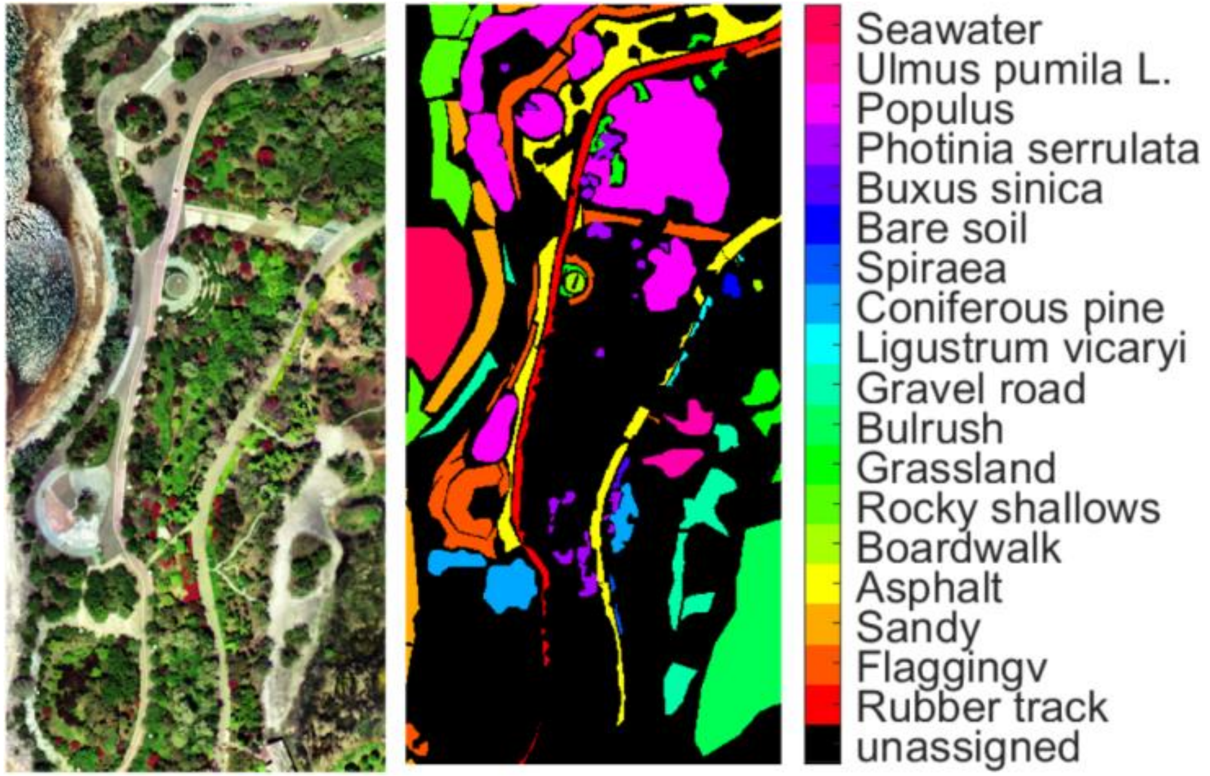
Methodology – Depthwise Attention Block



- The depthwise convolution to learn channel-wise spatial importance efficiently.
- Applies a sigmoid attention mask to enhance feature discrimination with low cost.

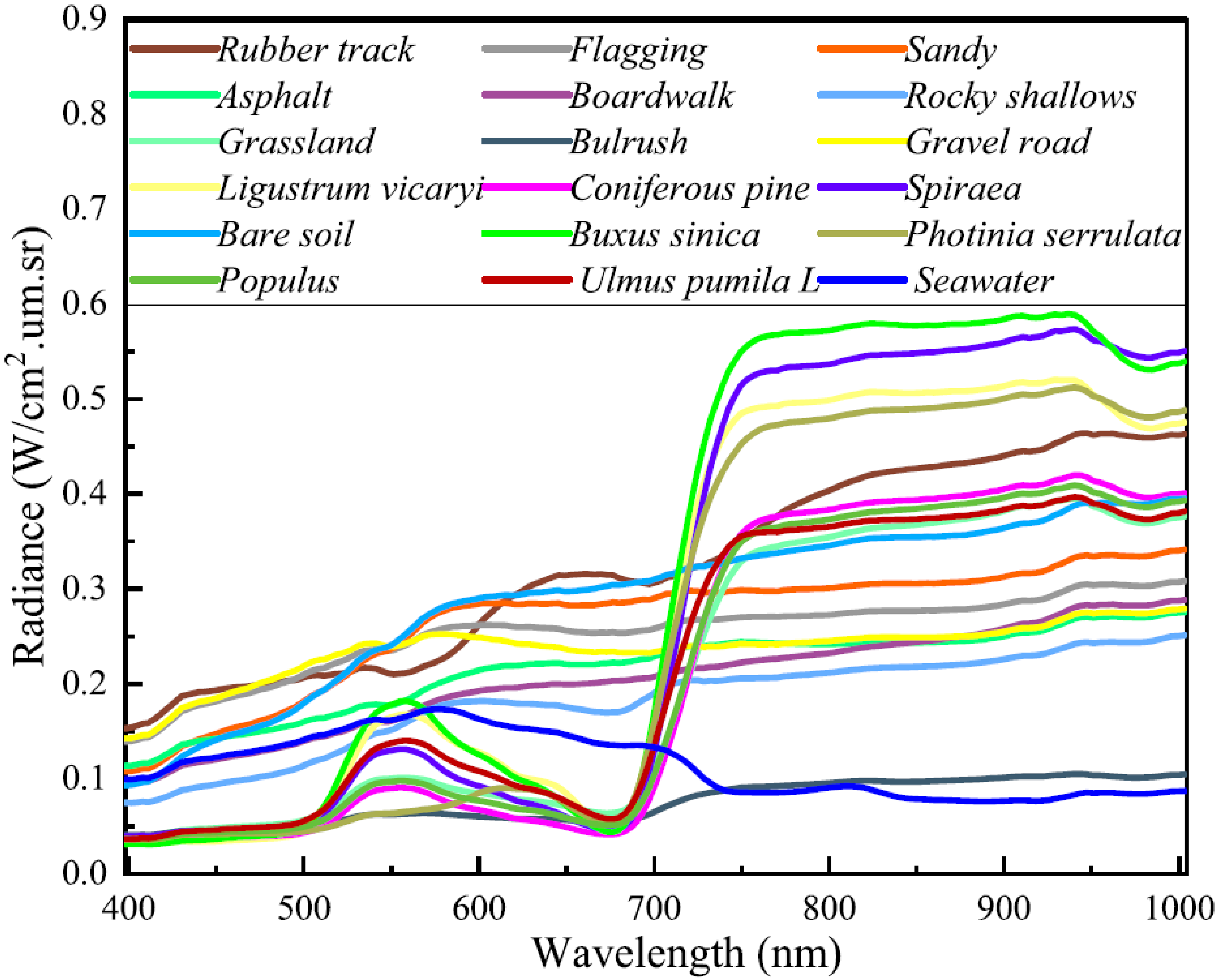
Dataset (QUH-Tangdaowan)

- Acquired on May 18, 2021. Gaiasky mini2-VN spectrometer
- 1740×860 with spatial resolution 0.15 m.
- 176 spectral bands ranges 400 nm – 1000 nm
- 557,366 labeled pixels
- **18 land-cover categories** including **vegetation types, artificial surfaces, and natural materials** such as *trees, grass, shrubs, soil, water, asphalt, gravel, and boardwalk areas*.



Fu, Hang, Genyun Sun, Li Zhang, Aizhu Zhang, Jinchang Ren, Xiuping Jia, and Feng Li. "Three-dimensional singular spectrum analysis for precise land cover classification from UAV-borne hyperspectral benchmark datasets." *ISPRS Journal of Photogrammetry and Remote Sensing* 203 (2023): 115-134.

Spectra of the Dataset Land-Cover Classes



Vegetation classes show a **strong red-edge rise near 700 nm**, indicating similar spectral behavior. **Non-vegetated surfaces** like soil, asphalt, and sand have **flatter curves**, while **seawater** shows the **lowest reflectance** across wavelengths.

Experiment Setup

- 1% for training , 1% for validation and the remaining are used for testing.
- **Epoch**: 100 epochs, early stopping strategy is adopted.
- **Batch size** 64.
- **Adam** optimizer with learning rate of **0.001.**
- **Categorical cross-entropy** as loss function.
- Implemented with Python, **Keras** framework that runs on top of **TensorFlow.**
- Patch size of 9x9 and number of PCs is set to 15.

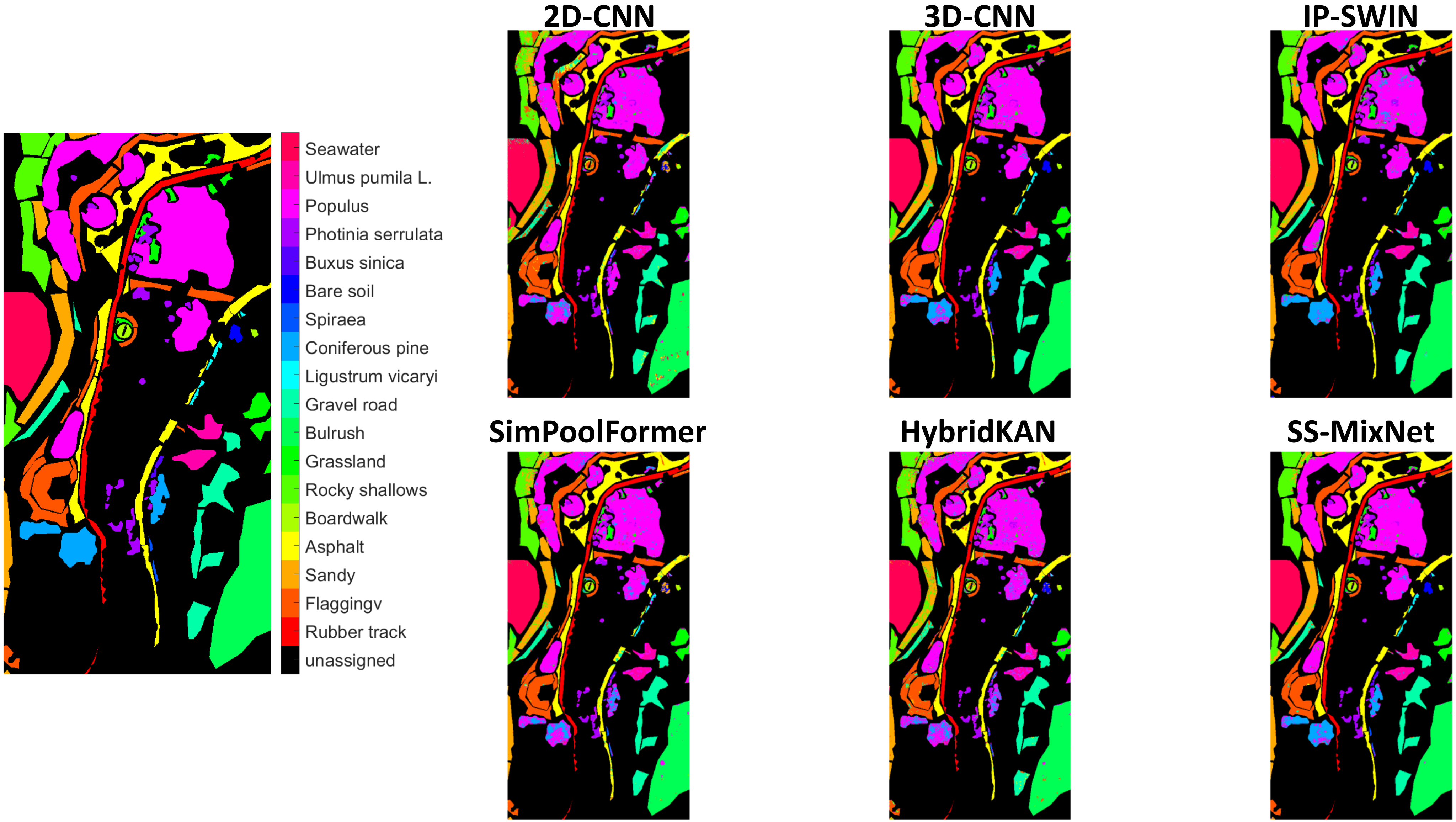
Results – Quantitative Comparison

Class	Train	Val	Test	2D-CNN	3D-CNN	IP-SWIN	SimPool Former	Hybrid KAN	SS-Mix Net
Rubber track	258	258	25,333	98.74	98.98	**99.83**	99.61	98.38	99.78
Flaggingv	555	555	54,443	90.25	**98.85**	98.34	97.16	92.91	98.45
Sandy	340	340	33,357	87.11	92.45	92.66	92.84	78.55	**95.08**
Asphalt	607	607	59,476	94.61	98.35	98.61	92.73	93.68	**99.18**
Boardwalk	19	19	1,824	35.18	**89.80**	83.46	87.54	71.00	86.31
Rocky shallows	371	371	36,383	64.78	90.07	88.16	83.53	87.07	**91.40**
Grassland	141	141	13,845	63.35	77.38	77.39	72.45	62.21	**77.94**
Bulrush	641	641	62,805	96.38	99.86	99.48	98.40	98.57	**99.80**
Gravel road	307	307	30,081	90.11	97.54	98.34	97.81	93.85	**98.54**
Ligustrum vicaryi	18	18	1,747	55.36	95.96	**98.93**	76.28	80.15	96.52
Coniferous pine	212	212	20,812	26.80	74.88	**88.93**	43.64	34.86	81.35
Spiraea	8	8	733	67.16	68.89	18.83	73.30	50.87	**80.64**
Bare soil	17	17	1,652	47.86	**99.94**	99.88	41.28	86.36	99.64
Buxus sinica	9	9	868	69.19	82.28	86.91	72.23	42.44	**93.91**
Photinia serrulata	140	140	13,740	76.78	**85.66**	83.79	68.98	64.38	83.22
Populus	1,409	1,409	138,086	92.97	94.18	91.70	92.13	88.35	**95.80**
Ulmus pumila L	98	98	9,606	62.08	76.97	85.59	76.76	64.09	**86.30**
Seawater	423	423	41,429	97.63	99.93	99.64	99.60	99.38	**99.98**
OA (%)				86.75	94.50	94.32	90.64	87.54	**95.68**
AA (%)				73.13	90.11	88.36	81.46	77.06	**92.44**
Kappa ($\times$100)				84.80	93.73	93.54	89.29	85.75	**95.08**

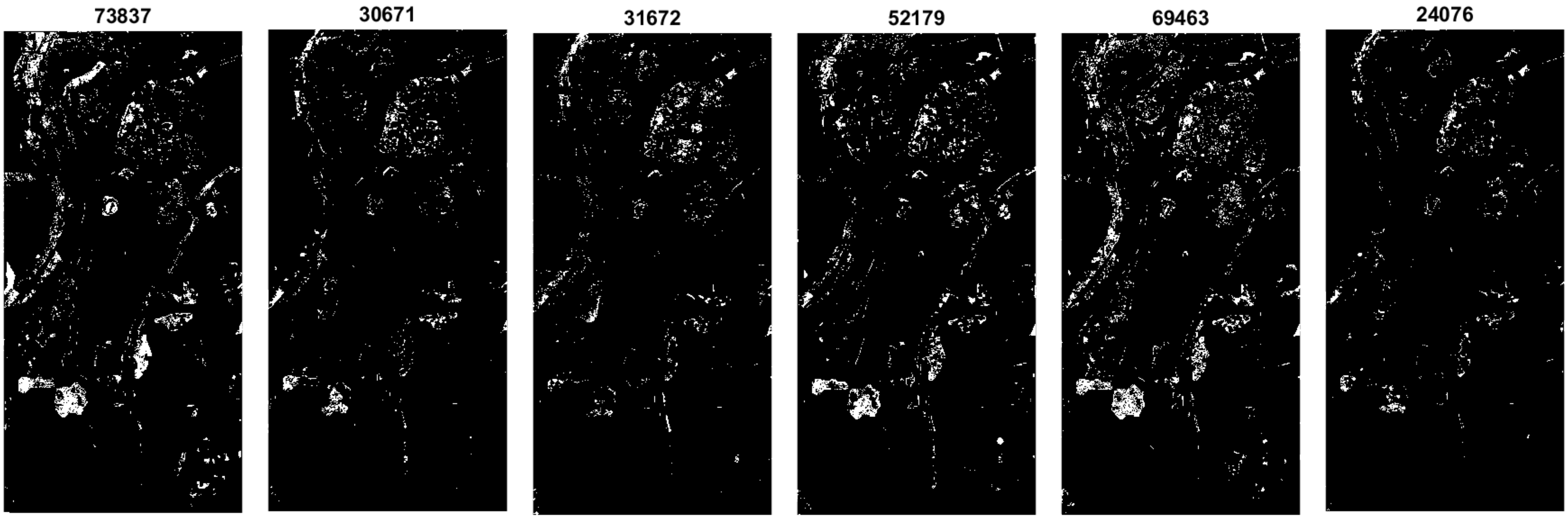
Results – Confusion Matrix

True \ Predicted	Rubber track	Flaggingv	Sandy	Asphalt	Boardwalk	Rocky shallows	Grassland	Bulrush	Gravel road	Ligustrum vicaryi	Coniferous pine	Spiraea	Bare soil	Buxus sinica	Photinia serrulata	Populus	Ulmus pumila L	Seawater
Rubber track	99.78	0.02	0.00	0.01	0.00	0.02	0.03	0.00	0.00	0.00	0.03	0.00	0.00	0.00	0.00	0.11	0.00	0.00
Flaggingv	0.00	98.45	0.01	0.45	0.00	0.09	0.00	0.01	0.43	0.00	0.12	0.00	0.00	0.00	0.00	0.35	0.09	0.00
Sandy	0.00	0.16	95.08	0.00	0.00	4.31	0.00	0.00	0.16	0.00	0.00	0.00	0.09	0.00	0.00	0.19	0.00	0.00
Asphalt	0.00	0.44	0.00	99.18	0.00	0.00	0.00	0.00	0.20	0.00	0.01	0.00	0.02	0.00	0.00	0.14	0.00	0.00
Boardwalk	0.00	8.54	0.00	0.97	86.31	0.11	0.00	0.00	0.11	0.00	0.00	0.00	2.95	0.00	0.00	0.97	0.05	0.00
Rocky shallows	0.00	0.23	7.68	0.00	0.00	91.40	0.23	0.04	0.10	0.00	0.04	0.00	0.00	0.00	0.00	0.26	0.00	0.01
Grassland	0.01	0.34	0.00	0.01	0.00	0.08	77.94	0.69	0.08	0.00	0.74	0.00	0.00	0.01	0.45	18.94	0.03	0.67
Bulrush	0.00	0.00	0.00	0.00	0.00	0.00	0.01	99.80	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.19	0.00	0.00
Gravel road	0.00	1.14	0.14	0.09	0.00	0.02	0.00	0.00	98.54	0.00	0.02	0.00	0.00	0.00	0.00	0.04	0.00	0.00
Ligustrum vicaryi	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	96.52	0.39	0.00	0.00	0.11	0.00	1.07	1.91	0.00
Coniferous pine	0.00	0.27	0.00	0.00	0.00	0.08	0.49	0.00	0.00	0.00	81.35	0.32	0.00	0.00	0.25	16.04	1.20	0.00
Spiraea	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.27	80.64	0.00	0.00	0.00	19.09	0.00	0.00
Bare soil	0.00	0.00	0.36	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	99.64	0.00	0.00	0.00	0.00	0.00
Buxus sinica	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.90	0.11	0.00	0.00	93.91	0.00	2.60	2.48	0.00
Photinia serrulata	0.14	0.18	0.00	0.54	0.00	0.00	0.14	0.00	0.00	0.00	0.39	0.00	0.00	0.00	83.22	15.34	0.06	0.00
Populus	0.01	0.18	0.08	0.04	0.00	0.29	1.10	0.04	0.05	0.05	1.43	0.04	0.01	0.15	0.44	95.80	0.30	0.00
Ulmus pumila L	0.00	0.07	0.00	0.06	0.00	0.00	0.52	0.00	0.00	0.63	2.00	0.00	0.00	0.09	0.02	10.28	86.30	0.02
Seawater	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	99.98

Results – Qualitative Comparison



Results – Qualitative Comparison (Error Maps)



2D-CNN 3D-CNN IP-SWIN SimPoolFormer HybridKAN SS-MixNet

Results – Model Efficiency (Parameters)

Model	Parameters	FLOPs	MACs
2D-CNN	287,950	1,658,448	829,224
3D-CNN	397,586	682,240	341,120
IP-SWIN	90,154	1,985,489	988,288
SimPoolFormer	771,122	57,497,293	28,423,255
HybridKAN	142,690	20,200,395	10,027,719
SS-MixNet	140,914	1,932,831	771,408

Conclusions and Future Work

- The proposed SS-MixNet effectively integrates 3D convolution, spectral–spatial MLP mixers, and depthwise attention, achieving superior accuracy and efficiency on QUH datasets with only 1% labeled data.

- The architecture demonstrates a strong balance between model complexity and performance, confirming its suitability for resource-limited hyperspectral applications.

- Future work will explore model pruning and knowledge distillation to further reduce complexity and inference time while preserving accuracy.

Thank You!

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